The rattly test mass: how large impulsive motion of a released test mass corrupts a frequency-modulation OPD measurement, and a motion-robust estimator that survives it

Physics of in-band injection, demonstration on FM1 Day09 data, and a validated robust pipeline

HET IFO OPD analysis

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Abstract

In the frequency-modulation method a known single tone of the laser frequency at f_{mod} writes a phase tone on the differential interferometer phase whose amplitude is proportional to the residual optical path-length difference (OPD). When the test mass is released it becomes a free harmonic oscillator, typically resonating at a few hertz, and its motion is the science signal. This note is about the single worst thing that can then happen to the measurement: a **rattly** test mass, that is one whose motion is not a smooth pendulum swing but an impulsive, repeatedly re-excited ring-down. We show that such motion deposits broadband power directly inside the narrow demodulation band around f_{mod}, even though its mechanical fundamental sits far below f_{mod}. The reason is a large coupling asymmetry: the differential phase is $\varphi = \nu(t)\,\mathrm{OPD}(t)/c$, so genuine OPD motion couples to phase through the full optical frequency ν_0, whereas the wanted tone couples only through the tiny modulation depth $\delta\nu_{\mathrm{pk}}$. The in-band gain of motion over signal is $\nu_0/\delta\nu_{\mathrm{pk}} \approx 1.0 \times 10^6$, so an OPD-motion component of a few nanometres at f_{mod} already rivals the tone. We derive this effect, show it in the FM1 Day09 released acquisitions (the near-tone noise floor of a rattly system is $12\times$ that of a clean one), and explain why it drives the standard incoherent estimator to over-report the OPD. We then implement and validate a motion-robust pipeline: an absolute near-tone contamination floor and reliability flag, a matched-filter coherent projection at the clock-locked tone frequency, burst gating, and a high-resolution readout of the test-mass motion from the low-frequency phase. On synthetic data with a known OPD the robust estimator recovers the truth to within 10 % at contamination levels where the standard estimator is biased high by up to $2.0\times$. All numbers and figures are regenerated from the raw data by `make_figures.py`.

Contents

1 The phenomenon

A released test mass in the measurement interferometer is a nearly free harmonic oscillator. Its fundamental frequency is typically a few hertz, well below the modulation tone $f_{\text{mod}} = 95\,\text{Hz}$ used in the FM1 campaign. During a released run the mass is set in motion (by the release itself, by residual forces, or by ground vibration) and rings. Two qualitatively different kinds of ring occur, and they have completely different consequences for the OPD estimate:

1. A **clean** ring-down: a smooth, slowly decaying sinusoid at the mechanical fundamental. Its energy is concentrated at the fundamental and a few discrete harmonics, all far below f_{mod}, and it dies away quietly. As we show below, this does not bias the OPD estimate at all, even when the swing is large (tens of cycles of differential phase).

2. A **rattly** ring-down: an impulsive, repeatedly re-excited motion (stick-slip, mechanical chatter, knocks). Its time series is choppy rather than sinusoidal, and, crucially, its spectrum has a broadband tail that reaches all the way up to and past f_{mod}.

Figure 1 contrasts the two directly from FM1 Day09 data. Both test masses were released and both moved by a comparable amount at low frequency (peak-to-peak 64.3 cycles for the clean R1 case, 406.5 cycles for the rattly R2 case), so the difference that matters is not how far the mass swung but the spectral character of the motion.

2 Physics: why low-frequency motion lands in the tone band

2.1 The two ways motion enters the phase

The differential interferometer phase (in cycles) is set by the instantaneous optical frequency $\nu(t)$ times the instantaneous OPD, divided by the speed of light,

$$\varphi(t) = \frac{\nu(t)\,\text{OPD}(t)}{c}. \quad (1)$$

Write the OPD as a static part plus motion, and the laser frequency as its carrier plus the modulation,

$$\text{OPD}(t) = \text{OPD}_0 + \delta(t), \quad (2)$$
$$\nu(t) = \nu_0 + \delta\nu_{\text{pk}} \cos(2\pi f_{\text{mod}} t). \quad (3)$$

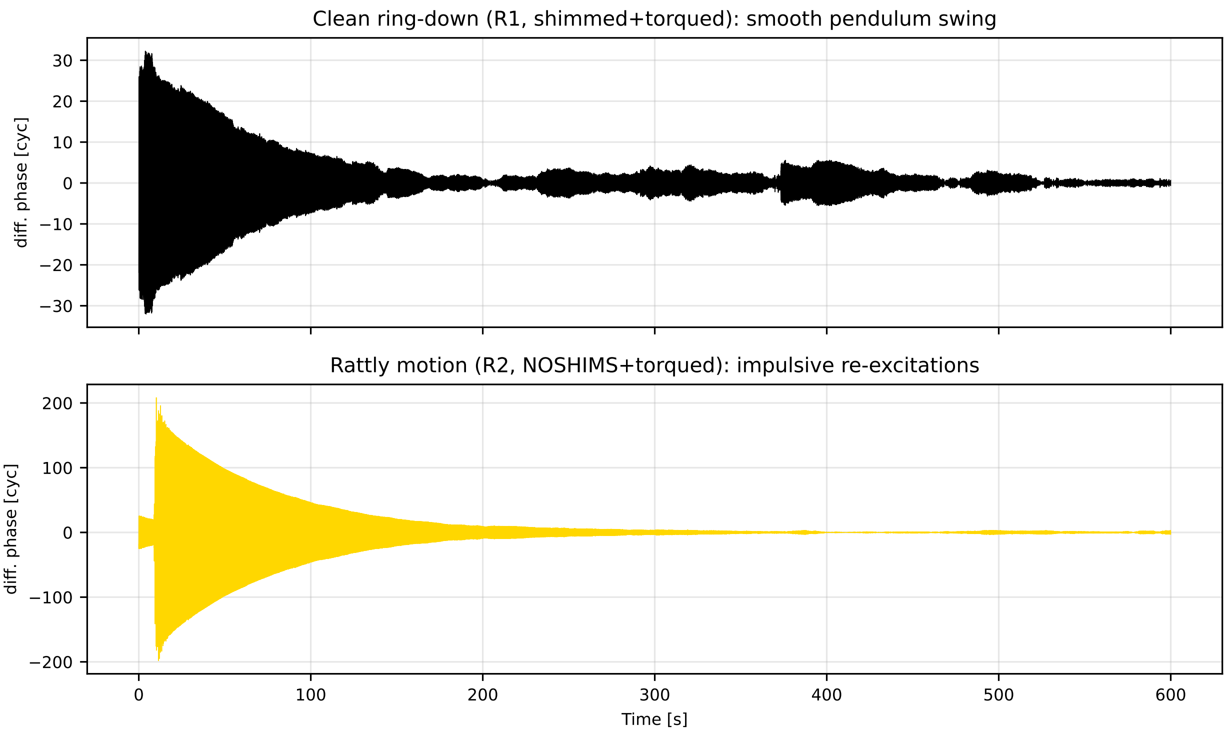

Figure 1: Raw differential phase of two released FM1 Day09 acquisitions. Top: a clean, smoothly decaying pendulum swing (system R1). Bottom: an impulsive, repeatedly re-excited motion (system R2). Both are large; only the second is dangerous, because only its motion has broadband power reaching the demodulation band.

Substituting into Eq. (1) and keeping the four products,

$$\varphi(t) = \underbrace{\frac{\nu_0\,\mathrm{OPD}_0}{c}}_{\mathrm{DC}} + \underbrace{\frac{\nu_0}{c}\,\delta(t)}_{\text{motion, gain } \nu_0/c} + \underbrace{\frac{\delta\nu_{\mathrm{pk}}}{c}\,\mathrm{OPD}_0\cos(2\pi f_{\mathrm{mod}}t)}_{\text{wanted tone, gain } \delta\nu_{\mathrm{pk}}/c} + \underbrace{\frac{\delta\nu_{\mathrm{pk}}}{c}\,\delta(t)\cos(2\pi f_{\mathrm{mod}}t)}_{\text{tone AM}}. \tag{4}$$

Two of these carry the motion $\delta(t)$, and it is essential to keep them apart.

The **tone-AM term** (last) is what intuition expects: it multiplies the motion by the tone, so a mechanical line at f_{mech} appears as sidebands at $f_{\mathrm{mod}} \pm f_{\mathrm{mech}}$. For $f_{\mathrm{mech}} \approx 7.7\,\mathrm{Hz}$ these sidebands sit at $95 \pm 7.7\,\mathrm{Hz}$, far outside the $\pm 0.5\,\mathrm{Hz}$ demodulation band, and are rejected. This term is not the problem.

The problem is the **second term**, $\nu_0\,\delta(t)/c$. It is the motion written straight onto the phase, with no shift in frequency. Any spectral content of $\delta(t)$ that genuinely lies at $f_{\mathrm{mod}} \pm 0.5\,\mathrm{Hz}$ lands directly inside the demodulation band. It cannot be filtered out by narrowing the band, because it is in the band.

2.2 The million-fold gain asymmetry

Compare the gains of the two motion-independent quantities we care about. The wanted tone enters with gain $\delta\nu_{\mathrm{pk}}/c$; the in-band motion enters with gain ν_0/c. Their ratio is

$$\frac{\nu_0}{\delta\nu_{\mathrm{pk}}} = \frac{1.934e+14\ \mathrm{Hz}}{188 \times 10^6\ \mathrm{Hz}} \approx 1.03e+06. \tag{5}$$

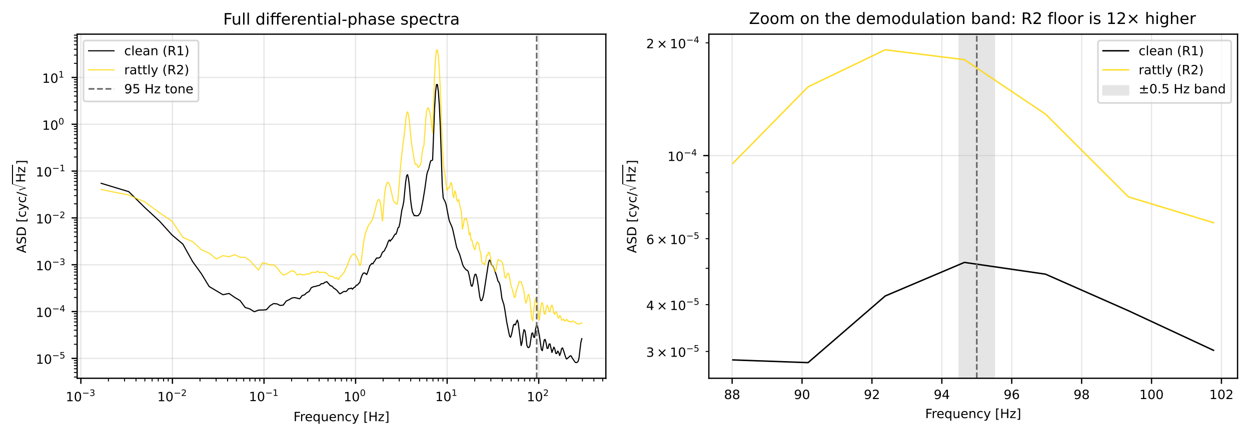

Figure 2: Differential-phase amplitude spectral density. Left: full band; both systems show the few-hertz mechanical motion and harmonics. Right: zoom on the demodulation band (grey), where the rattly system's broadband floor is 12× higher. The danger is set by the spectral flatness of the motion (impulsive versus sinusoidal), not by how far the mass swings.

Here $\nu_0 = c/\lambda$ with $\lambda = 1550\,\text{nm}$, and $\delta\nu_{\text{pk}}$ is the peak laser frequency deviation of the modulation. So a motion component at f_{mod} that is only about 10^{-6} of the low-frequency swing already rivals the signal tone. Numerically, a mere 2 nm of δOPD content at f_{mod} reads as about 2.1 mm of apparent OPD ($2\,\text{nm} \times 1.03e + 06$). This is the origin of the millimetre-scale spikes seen in contaminated records.

2.3 So the demodulator's leakage immunity is real but specific

The complex-baseband demodulator is immune to spectral leakage: it kills the sidelobe leakage of strong out-of-band lines, so the large 7.7 Hz fundamental and its low harmonics do not leak in through the discrete-Fourier-transform window. That immunity is genuine and is why a clean ring-down (Section 1, case 1) is harmless however large. But no filter can remove power that truly lies at $f_{\text{mod}} \pm 0.5\,\text{Hz}$. A rattly motion has exactly that: a broadband tail with real power in the band, amplified by Eq. (5).

3 Evidence in the FM1 Day09 data

3.1 The near-tone floor, not the swing amplitude

Figure 2 shows the differential-phase spectra of the same clean and rattly acquisitions. On the full spectrum both show the mechanical fundamental and a harmonic comb. The decisive difference is local to the tone: zooming on the demodulation band, the rattly system has a broadband floor 12× higher than the clean one ($1.62e-04$ versus $1.35e-05$ cyc/$\sqrt{\text{Hz}}$ in the $f_{\text{mod}} \pm [0.6, 1.5]\,\text{Hz}$ flanks). This elevated in-band floor, amplified by the 10^6 factor of Eq. (5), is what floods the estimator.

3.2 How the standard estimator is fooled

The primary pipeline estimator (see the companion note on OPD estimation) demodulates the tone to a complex baseband phasor $z(t)$ whose magnitude is the instantaneous tone amplitude. It then chooses between a coherent estimate (the vector mean $|\langle z\rangle|$, matched to a phase-stable tone) and

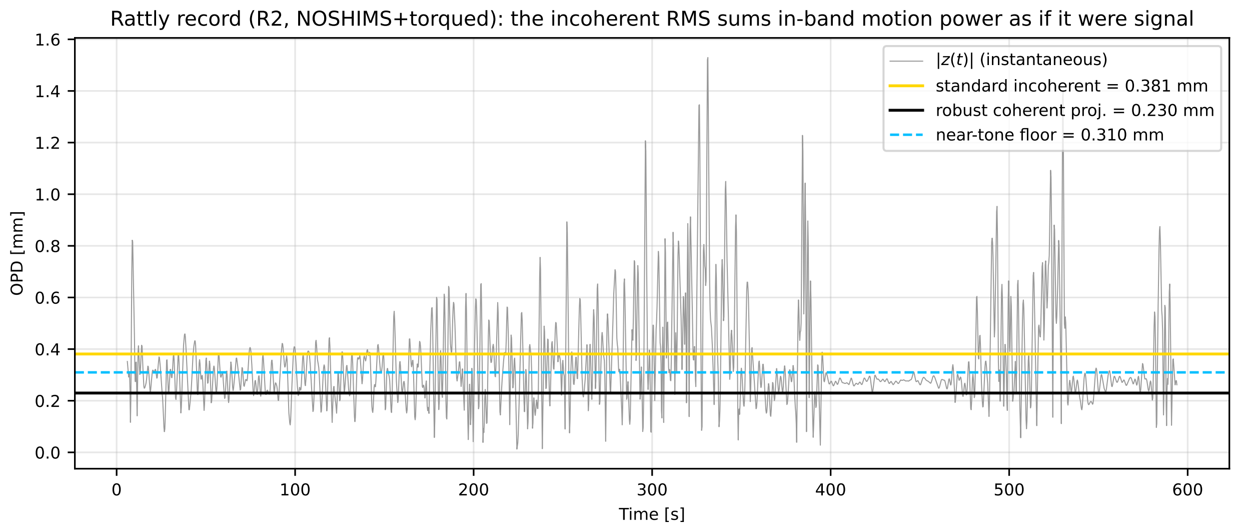

Figure 3: Anatomy of a rattly record. The instantaneous tone magnitude $|z(t)|$ (grey) is inflated by in-band motion power. The standard incoherent estimate (red) sums that power as signal; the matched-filter coherent projection (blue) rejects the zero-mean part and tracks the lower baseline; the near-tone floor (green dashed) shows how large the in-band noise is relative to the tone.

an incoherent estimate (the noise-debiased root-mean-square $\sqrt{\langle |z|^2 \rangle - S_n}$, matched to a wandering tone), based on a measured coherence. In-band contamination breaks this logic in two ways:

1. It adds random-phase power to $z(t)$, which lowers the measured coherence. The selector reads this as a wandering tone and switches to the incoherent branch, even though the static OPD tone is not actually wandering.
2. The incoherent branch then reports $\sqrt{\langle |z|^2 \rangle - S_n}$. Since the contamination adds to $\langle |z|^2 \rangle$ and the single far off-tone reference under-estimates the coloured near-tone floor S_n, the branch sums the in-band motion power as if it were signal and over-reports the OPD.

Figure 3 shows this for the genuinely rattly Day09 R2 record (NOSHIMS, torqued, RMS motion 46 µm). The instantaneous $|z(t)|$ wanders far above the coherent baseline; the standard incoherent estimate (0.381 mm) sits high in that excursion, whereas the coherent projection (0.230 mm) tracks the lower baseline, and the near-tone floor (310 µm-equivalent) shows the in-band noise is comparable to the tone itself. The separate R2 record that originally stood out with an anomalously large OPD (0.840 mm) is an even more extreme case: its near-tone floor is 408 µm-equivalent, comparable to the apparent signal, so it is flagged (reliability 2.2) rather than trusted.

4 A motion-robust pipeline

The physics of Section 2 points to four concrete defences, implemented in `het_ifo_opd.estimators.demodulate_` and exposed through `estimate_opd_robust`.

4.1 An absolute near-tone contamination floor and a reliability flag

Rather than one far off-tone reference at $f_{\mathrm{mod}} + 7\,\mathrm{Hz}$, the noise floor is measured from several bins immediately flanking the tone (at $f_{\mathrm{mod}} \pm \{1.2, 1.6, 2.0\}\,\mathrm{Hz}$, whose passbands clear the tone) and combined by a median. This gives the true coloured floor next to the tone, S_n^{adj}. Two diagnostics follow: a relative contamination ratio $S_n^{\mathrm{adj}}/S_n^{\mathrm{far}}$, and, more usefully, an absolute reliability

$$\rho_{\mathrm{rel}} = \frac{A_{\mathrm{robust}}}{\sqrt{S_n^{\mathrm{adj}}}}, \tag{6}$$

the ratio of the recovered tone amplitude to the near-tone noise amplitude. A record with $\rho_{\mathrm{rel}} \gtrsim 3$ is trustworthy; a smaller value flags a record where the motion-driven in-band noise is comparable to the tone and the OPD cannot be certified from that acquisition.

4.2 A matched-filter coherent projection at the clock-locked tone

The static OPD produces a strictly constant phasor at exactly f_{mod} (the drive and the phasemeter share a clock), so its optimal estimator is the vector mean

$$A_{\mathrm{robust}} = \left|\langle z(t)\rangle\right|. \tag{7}$$

This is the matched filter for a constant, clock-locked tone. Any zero-mean in-band contamination, which is what broadband motion power at f_{mod} looks like, averages to zero in the vector mean, so the estimate is unbiased however large that contamination is. Two points are important. First, the estimator does not switch to the incoherent branch under low coherence, because here the low coherence is contamination, not tone wander. Second, it deliberately avoids the data-driven linear-phase-ramp de-rotation used by the default demodulator: a phase corrupted by contamination corrupts that fit and would smear the true tone. The tone frequency is fixed once, robustly, by the clock-locked refinement.

4.3 Burst gating

Rattly contamination arrives in bursts. An in-band noise witness (the magnitude of an adjacent-band demodulation versus time) marks the quiet epochs, and the coherent projection is formed over those epochs. When quiet epochs exist (as they do between re-excitations of a ring-down) this sharpens the estimate and reduces residual bias; when they do not, the reliability flag catches the record.

4.4 The test-mass motion as a bonus

Finally, the same physics that makes the in-band tail dangerous makes the low-frequency phase an exquisite motion sensor. The second term of Eq. (4) gives the displacement directly,

$$\delta(t) = \frac{c}{\nu_0}\,\varphi_{\mathrm{LF}}(t) = \lambda\,\varphi_{\mathrm{LF}}(t), \tag{8}$$

where φ_{LF} is the band-limited low-frequency differential phase in cycles and one cycle equals one carrier wavelength of OPD. Laser frequency noise is common-mode-rejected in the difference, so this is a clean, high bandwidth readout of the released mass, reported alongside the OPD (Figure 4).

5 Validation on synthetic data with a known OPD

Real records have no ground truth, so we validate the estimator against a known injected OPD. The synthetic differential phase is a constant tone at f_{mod} encoding $\mathrm{OPD}_0 = 0.300\,\mathrm{mm}$, plus a large

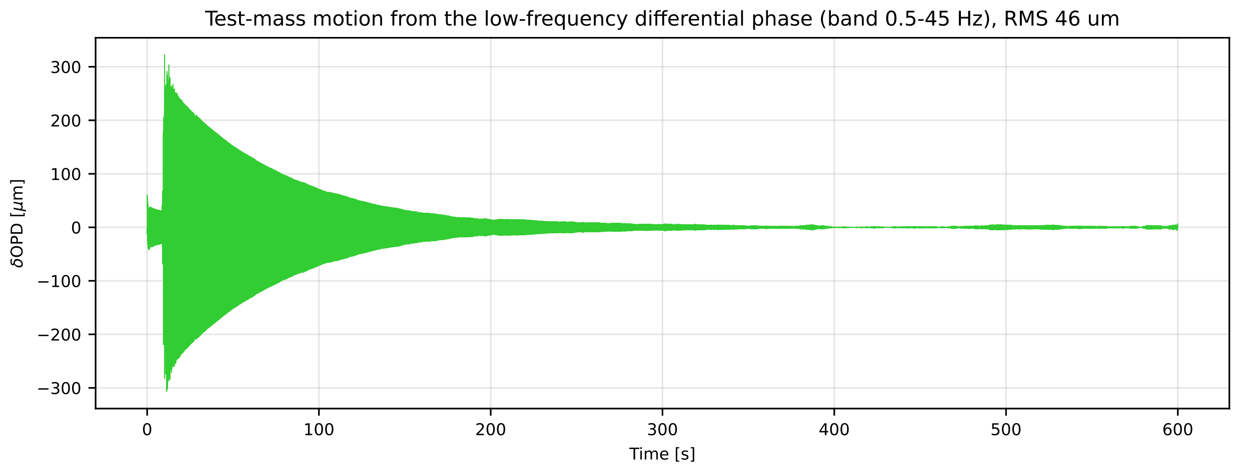

Figure 4: Test-mass motion recovered from the low-frequency differential phase of the rattly R2 acquisition. This is the force-sensing signal; the same in-band tail of this motion is what contaminates the tone.

low-frequency motion of several cycles, plus a controlled random-phase narrowband contamination centred on f_{mod} (this is exactly broadband motion power landing in the band with random phase), plus white noise. We sweep the contamination level from zero to four times the tone amplitude, in a bursty and in a steady variant.

Figure 5 is the central result. The standard estimator tracks the truth at low contamination but, once the contamination exceeds the tone, it switches to the incoherent branch and climbs well above the truth, reaching 0.607 mm (bursty) and 0.588 mm (steady) at four times the tone, that is up to 2.0× the true OPD. The robust estimator stays on the truth throughout, recovering 0.301 mm and 0.270 mm respectively, and within 10 % of the truth across the whole sweep. This is the sense in which the estimator gets much better at recovering the true OPD in the presence of large test-mass motion.

6 The FM1 Day09 released acquisitions, before and after

Table 1 runs both pipelines on every Day09 released file. The pattern is systematic. The R1 system rang cleanly: all four records are coherent, their near-tone floors are small (14 to 37 µm-equivalent), their reliabilities are well above 3, and the robust and standard OPDs agree. The R2 system rang rattly: every record is pushed to the incoherent branch by in-band contamination, the near-tone floors are an order of magnitude larger (132 to 408 µm-equivalent), and three of the four are flagged as low-reliability. For the genuinely rattly R2 record (NOSHIMS, torqued, RMS motion 46 µm) the standard estimate of 0.381 mm collapses to a flagged 0.230 mm once the summed motion power is removed. The R2 record that originally stood out as the large-OPD outlier is not silently corrected but correctly flagged (reliability 2.2), signalling that its OPD is not trustworthy and the mass should be damped and the run repeated.

Figure 6 shows the same comparison graphically.

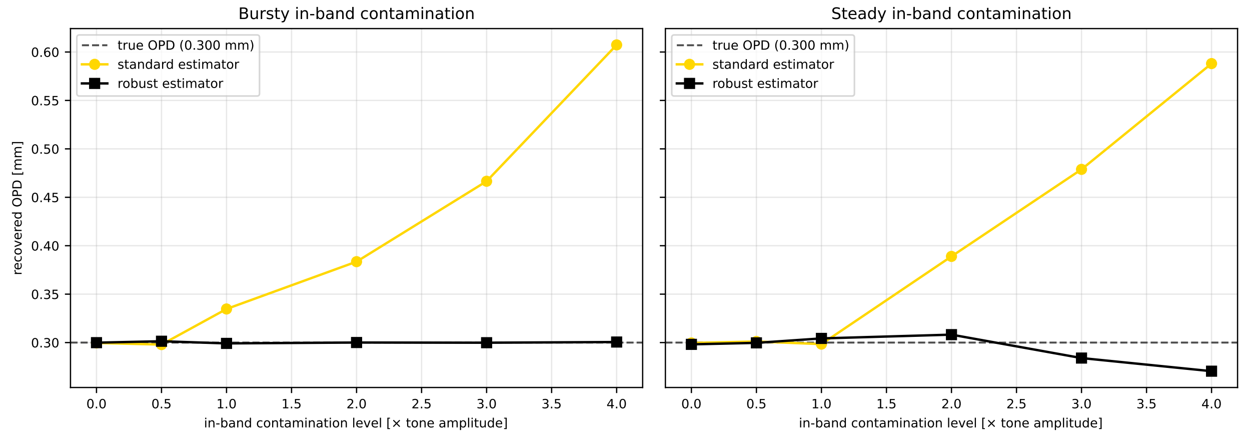

Figure 5: Recovered OPD versus in-band contamination level, for a known true OPD of 0.300 mm (dashed). Left: bursty contamination; right: steady. The standard estimator (red) is driven high once the contamination exceeds the tone; the robust estimator (blue) stays on the truth.

Table 1: FM1 Day09 released acquisitions: standard versus robust OPD. Floor is the absolute near-tone noise (µm-equivalent OPD); reliability is amplitude over floor; motion is the RMS test-mass displacement from the low-frequency phase. A dagger marks robust OPDs flagged as low-reliability ($\rho_{\mathrm{rel}} < 3$).

Sys	Config	$\mathrm{OPD}_{\mathrm{std}}$ [mm]	mode	$\mathrm{OPD}_{\mathrm{robust}}$ [mm]	floor [µm]	ρ_{rel}	motion [µm]
R1	NOSHIMS	0.241	coherent	0.206	14	14.3	2
R1	NOSHIMS+torqued	0.127	coherent	0.115	37	3.1	13
R1	shimmed+torqued	0.288	coherent	0.291	31	9.3	9
R1	shimmed+torqued (vac)	0.089	coherent	0.080	25	3.3	22
R2	NOSHIMS	0.261	incoherent	0.459	132	3.5	3
R2	NOSHIMS+torqued	0.381	incoherent	$0.230^{\dagger}$	310	0.7	46
R2	shimmed+torqued	0.840	incoherent	$0.883^{\dagger}$	408	2.2	5
R2	shimmed+torqued (vac)	0.271	incoherent	$0.209^{\dagger}$	304	0.7	9

7 Recommendations

Experimental. The cheapest and most effective fix is upstream: damp the test mass before acquiring, and avoid impulsive re-excitation (gentle release, isolate from knocks and stick-slip). A large but clean ring-down is harmless; a small but rattly one is not. Where practical, record long so that quiet epochs between re-excitations can be selected in analysis, and record the mechanical spectrum so the broadband tail can be seen directly.

Analysis. Use the robust pipeline for released runs. Always report the near-tone floor and the reliability flag next to the OPD, and treat a flagged record as an upper limit rather than a measurement. Report the recovered test-mass motion as the force-sensing signal in its own right.

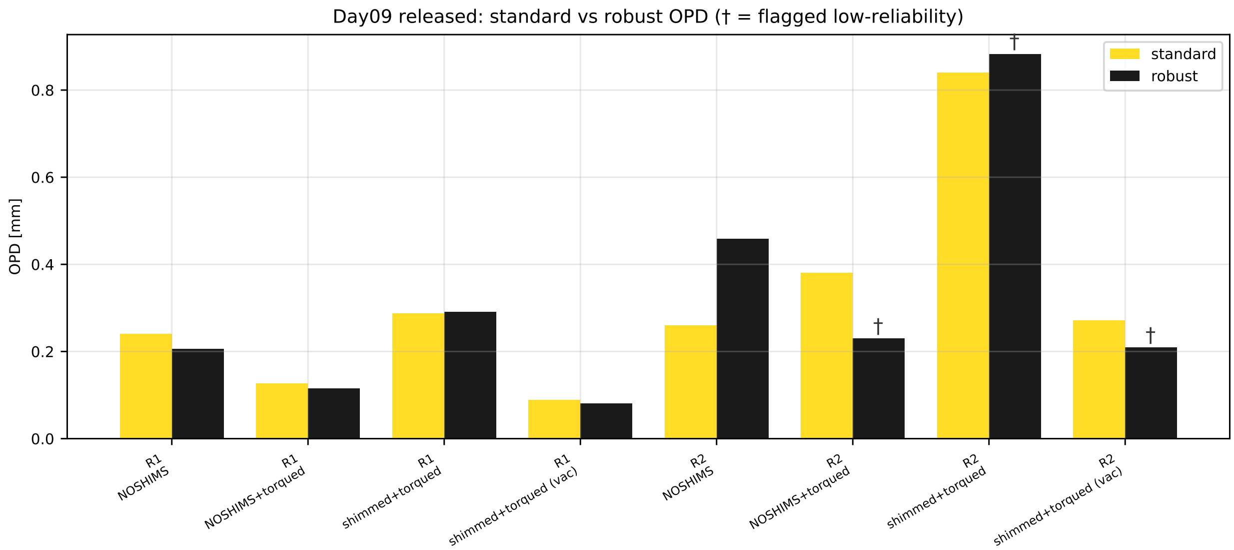

Figure 6: Standard (red) versus robust (blue) OPD for every Day09 released acquisition. The R1 system is unchanged; the R2 system is where the robust estimator and the reliability flag (†) do their work.

8 Reproducibility

Every number and figure in this note is regenerated from the raw FM1 phasemeter files by `make_figures.py`, which runs the standard (`estimate_opd`) and robust (`estimate_opd_robust`) pipelines on the Day09 acquisitions and the synthetic validation, and writes the figures, `values.tex`, and `table_day09.tex` that this document includes. Build with `make` in this directory.